$$\hat{y} = b_0 + b_1 x_1$$

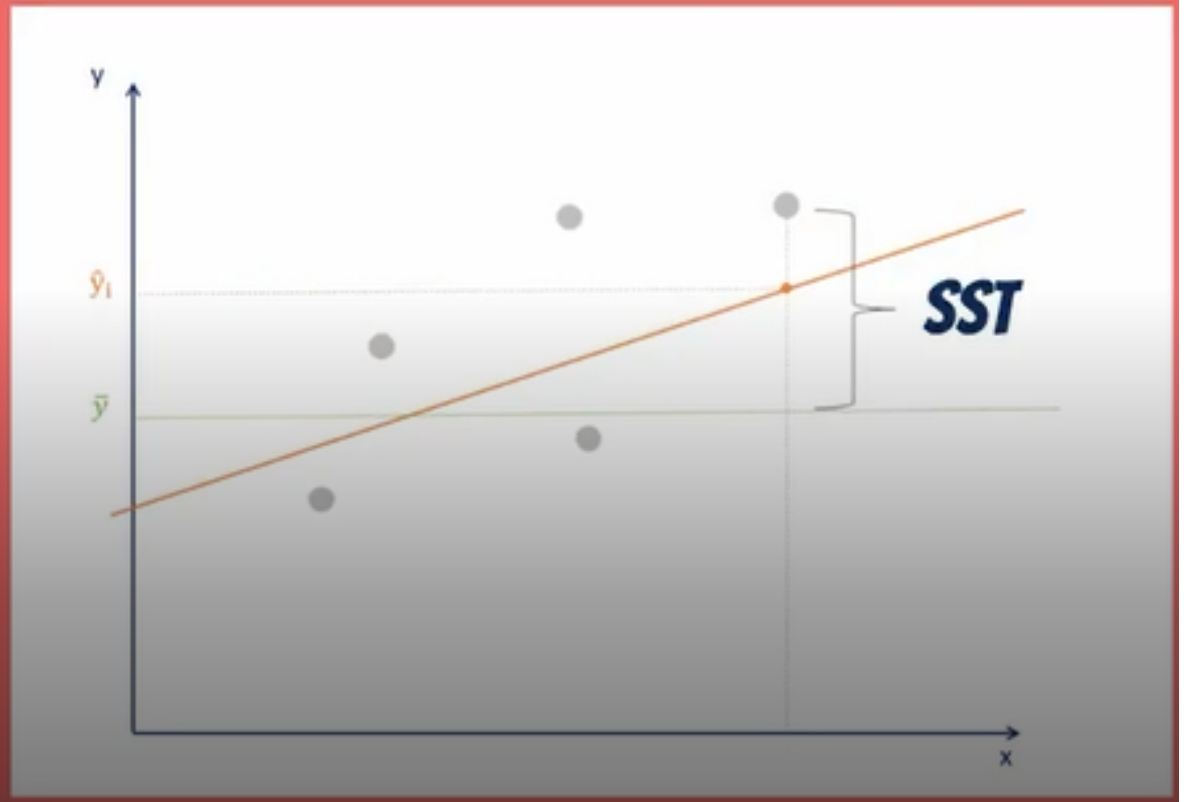
College GPA =0.48 + 0.0015*SAT

SST

Sum of squares total

Measures the total variablity of the dataset

$$\sum_{i=1}^{n}(y_i - \bar{y})^2$$

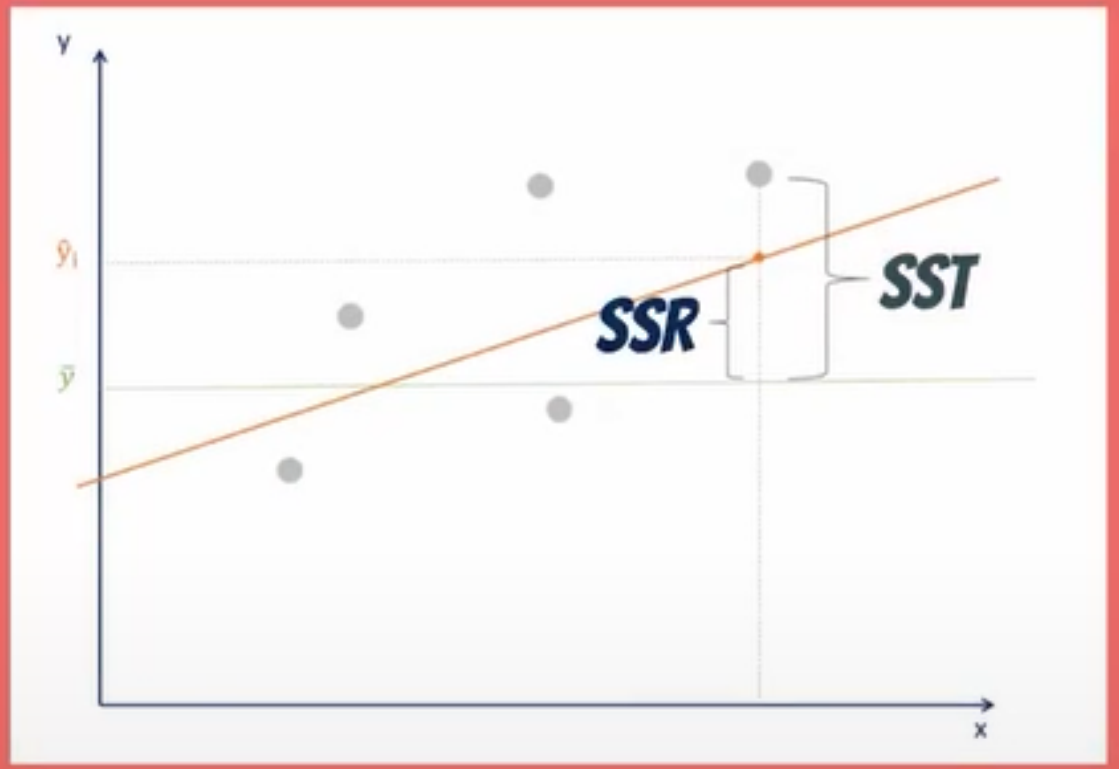


SSR

Sum of squares regression

Measures the explained variablity by your line

$$\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$



SSE

Sum of squares error

Measures the unexplained variablity by the regression

$$\sum_{i=1}^{n} e_i^2$$

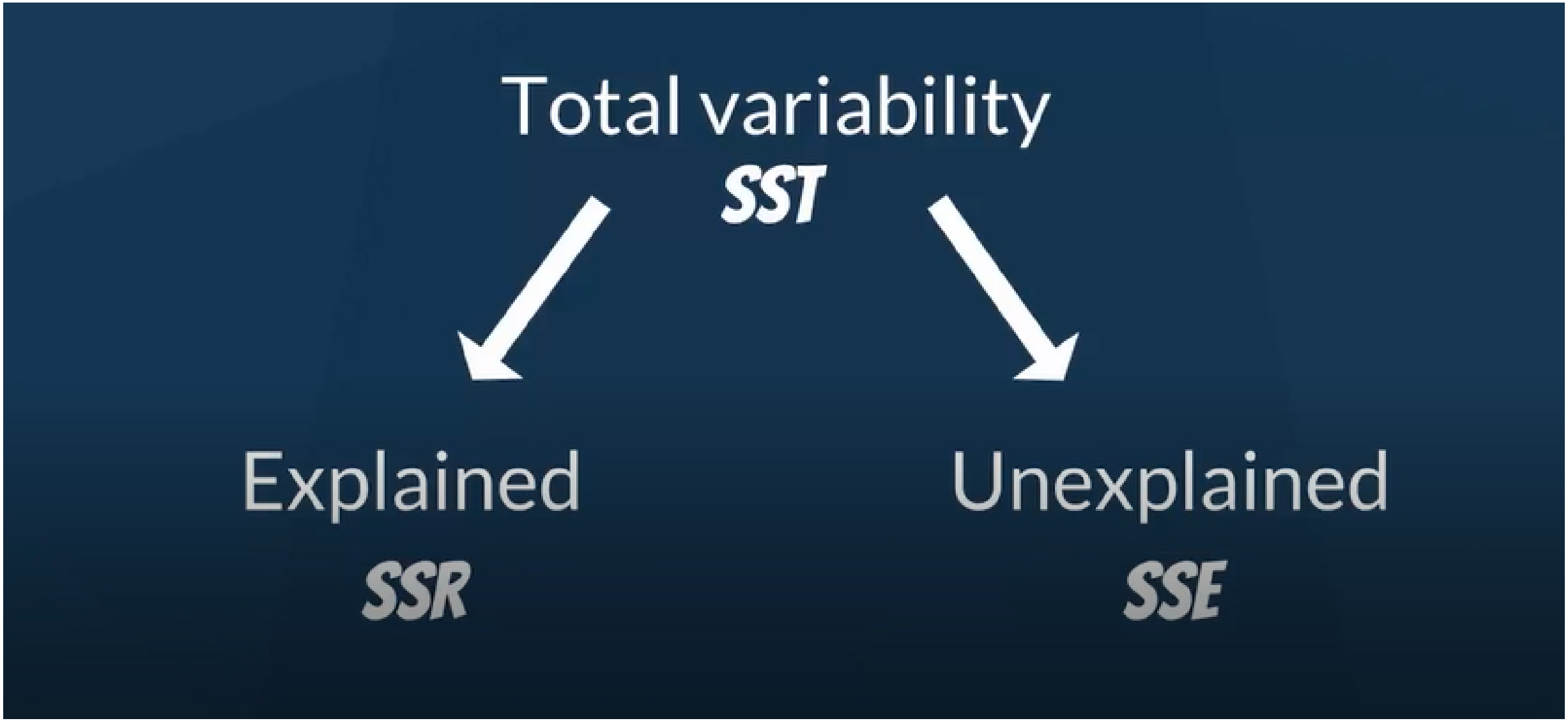


Connection?

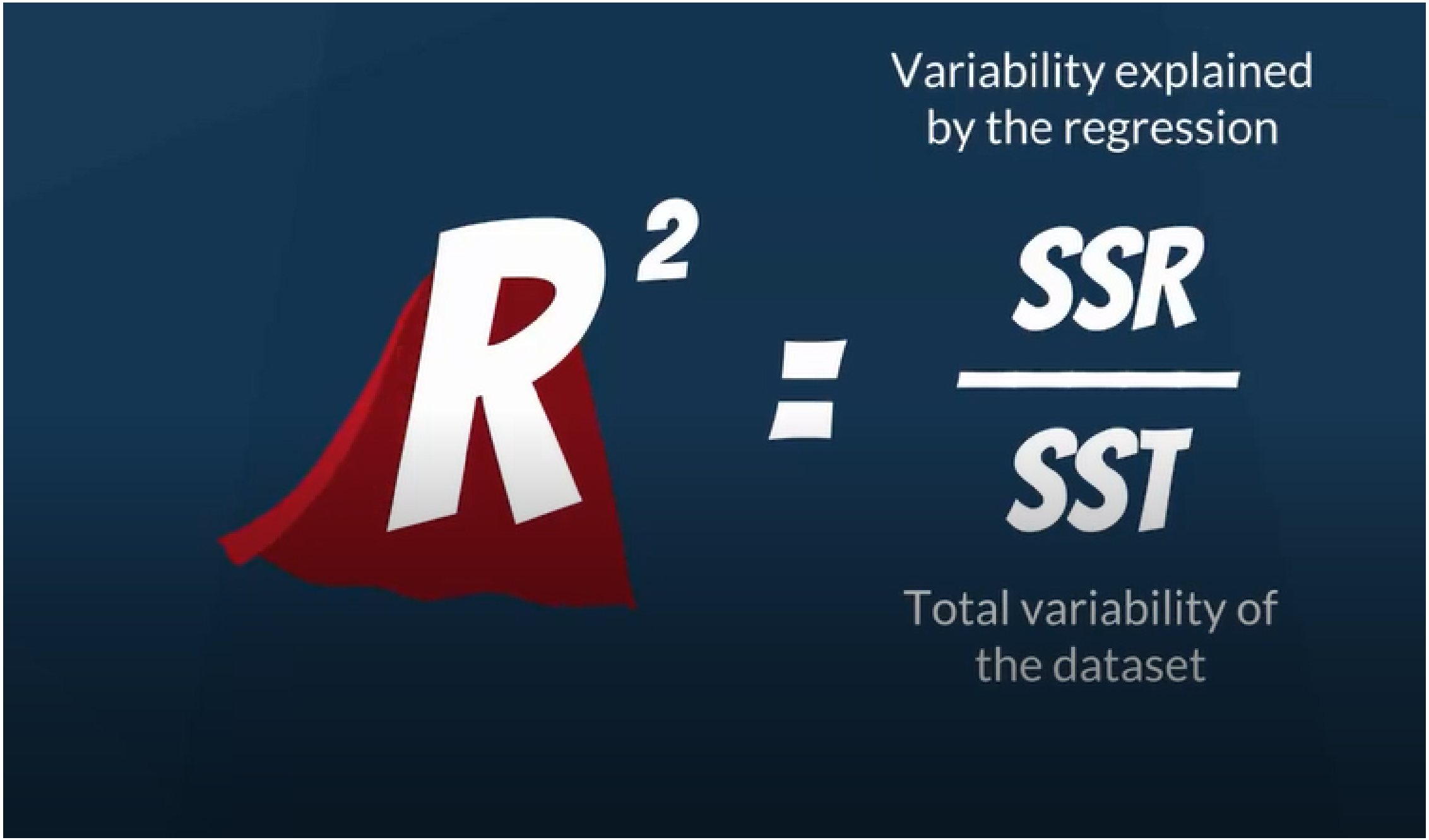
SST = SSR + SSE

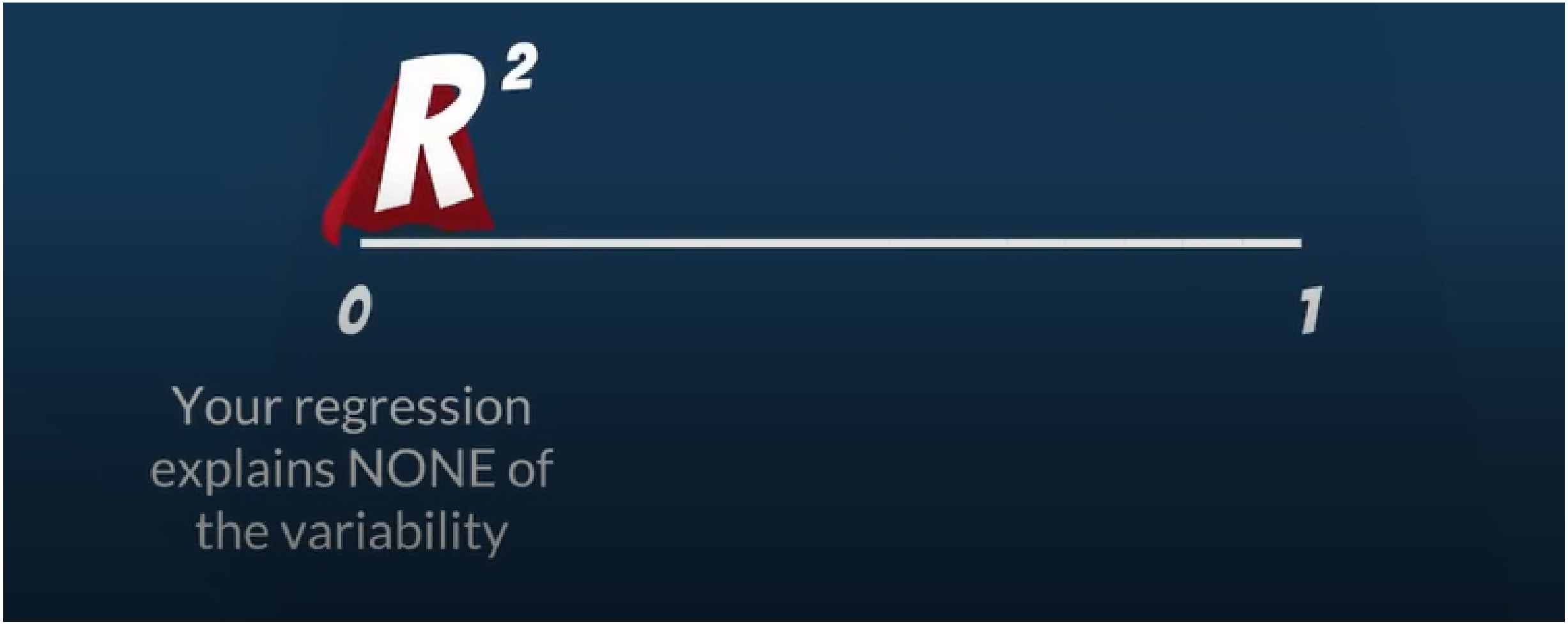
$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n} e_i^2$$

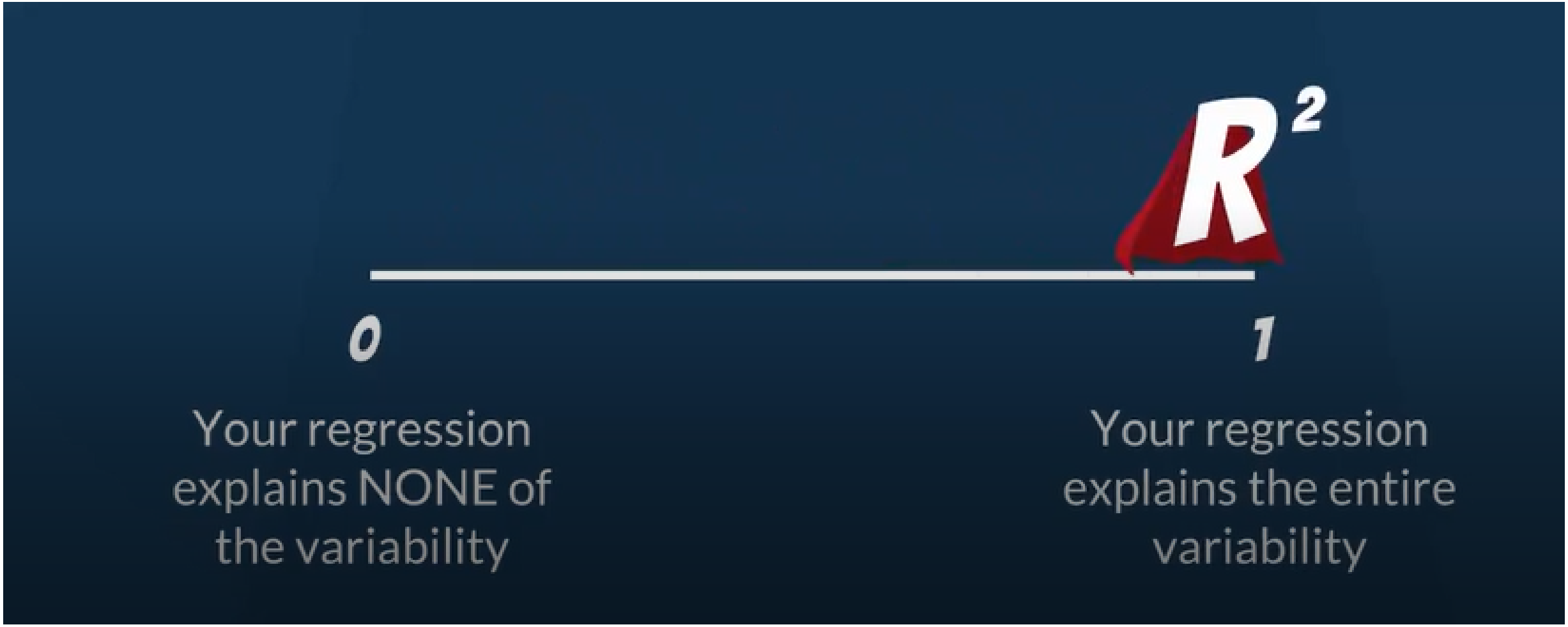


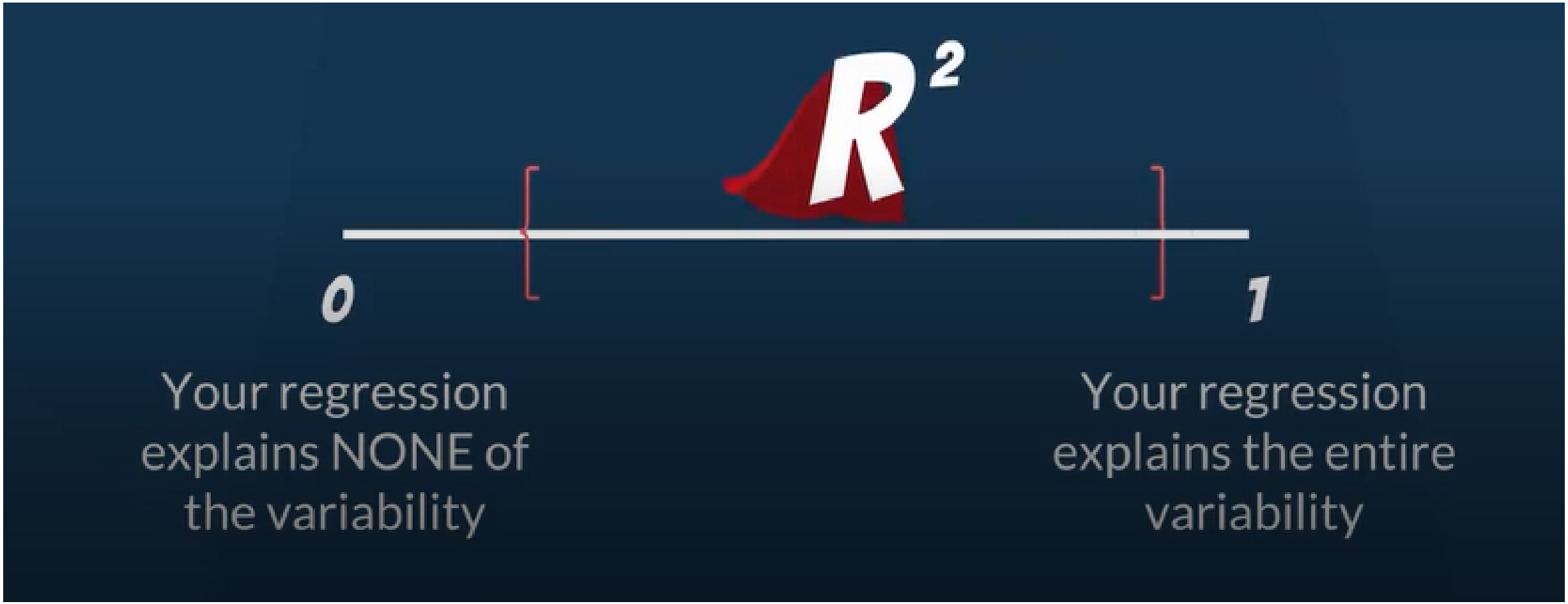

Total variability
SST
Explained
SSR
Unexplained
SSE

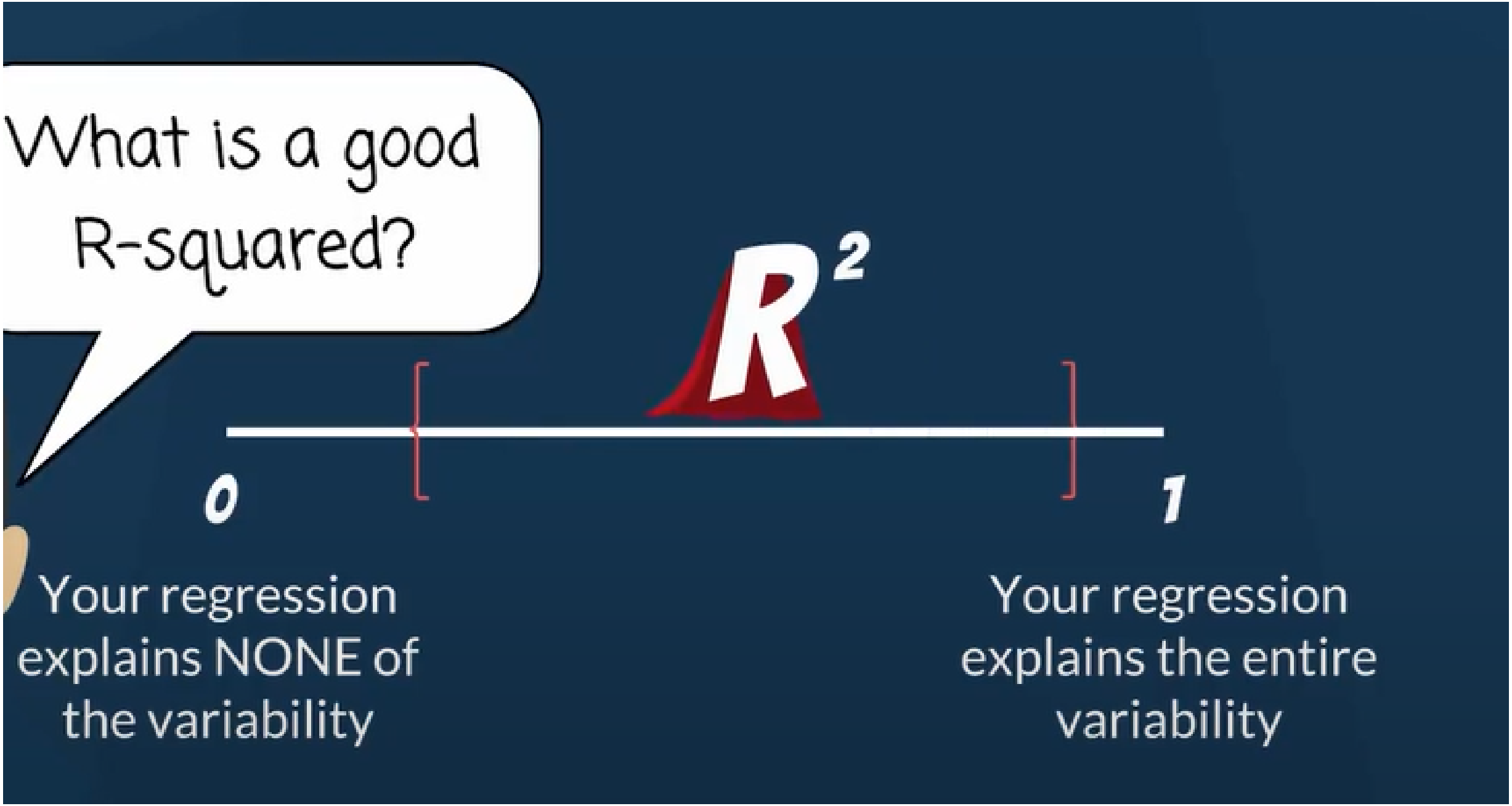

R 2

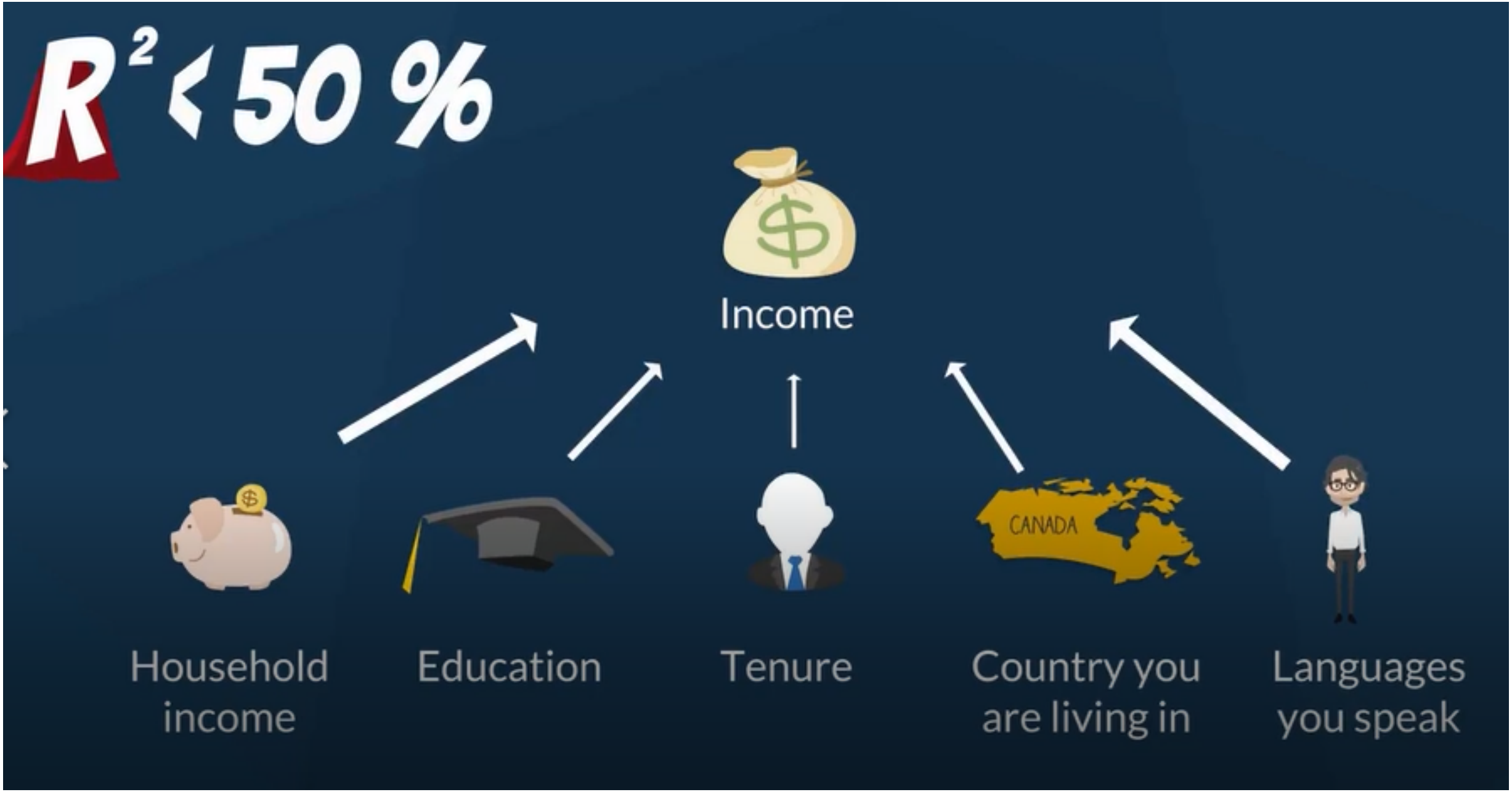

Variability explained
by the regression
$R^2 = \frac{SSR}{SST}$
Total variability of
the dataset

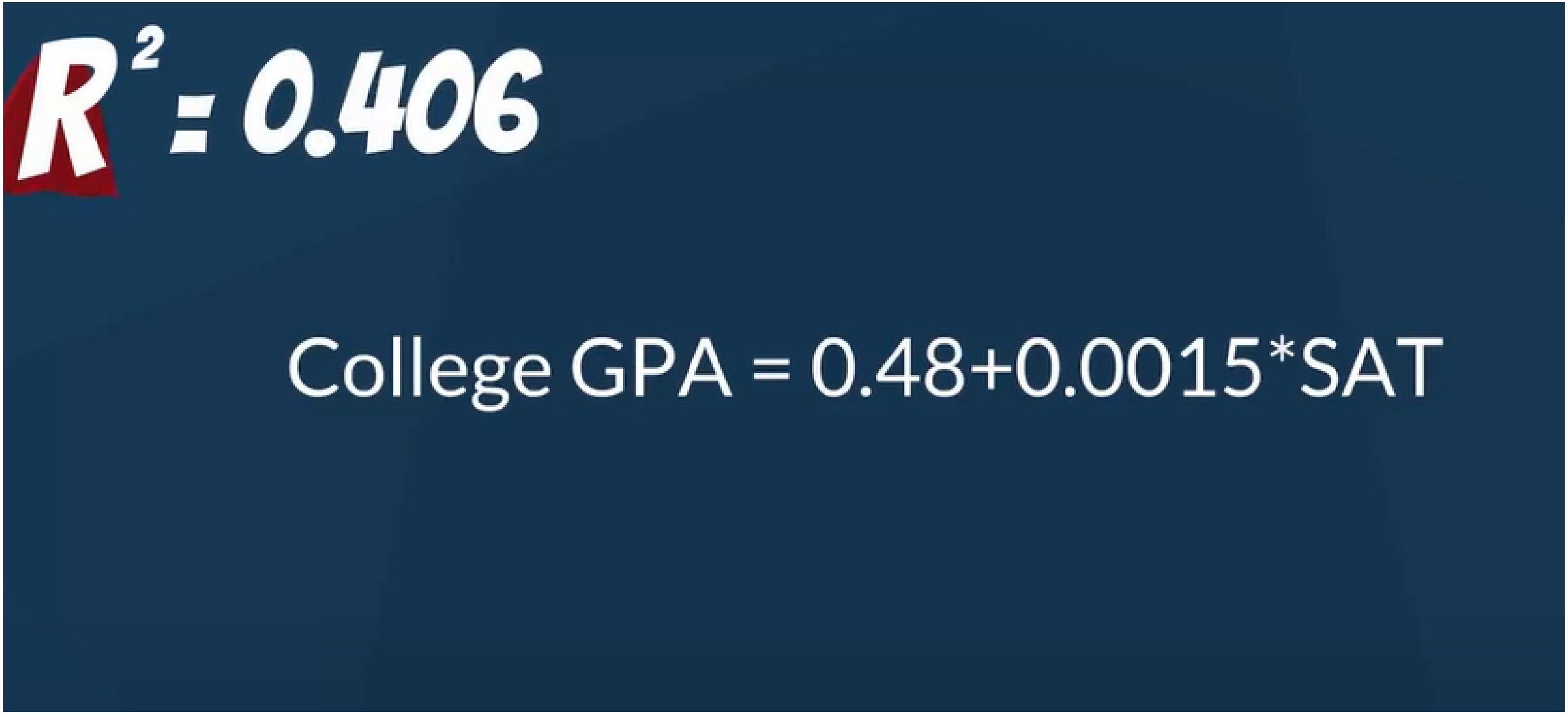

R^2
0
1
Your regression explains NONE of the variability


R^2
0
1
Your regression explains NONE of the variability
Your regression explains the entire variability


R^2
0
1
Your regression explains NONE of the variability
Your regression explains the entire variability


What is a good R-squared?
R^2
0
1
Your regression explains NONE of the variability
Your regression explains the entire variability


$R^2 < 50\%$
Income
Household income
Education
Tenure
Country you are living in
Languages you speak
CANADA


$R^2 = 0.406$
College GPA = 0.48+0.0015*SAT

OLS

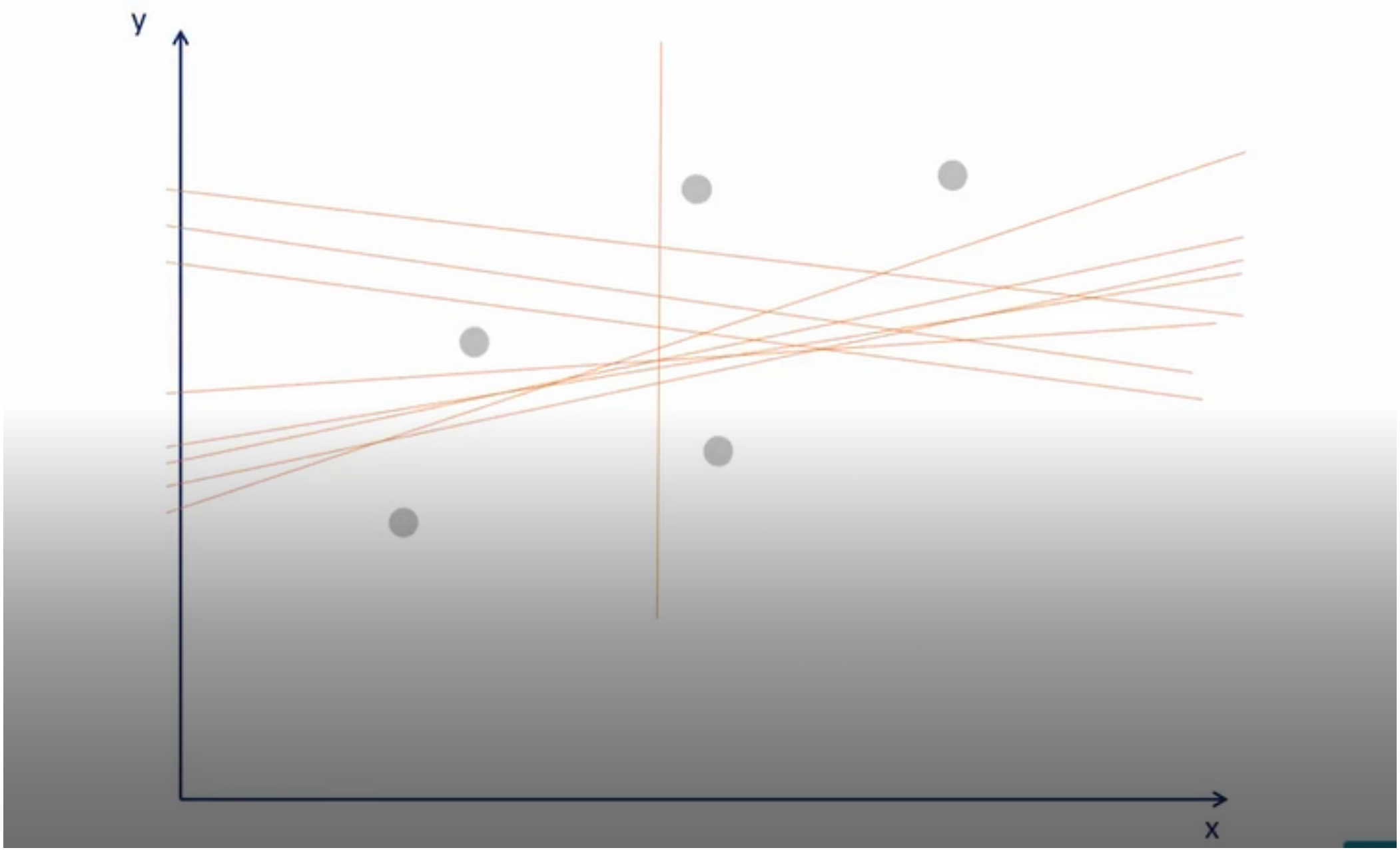
Ordinary least squares

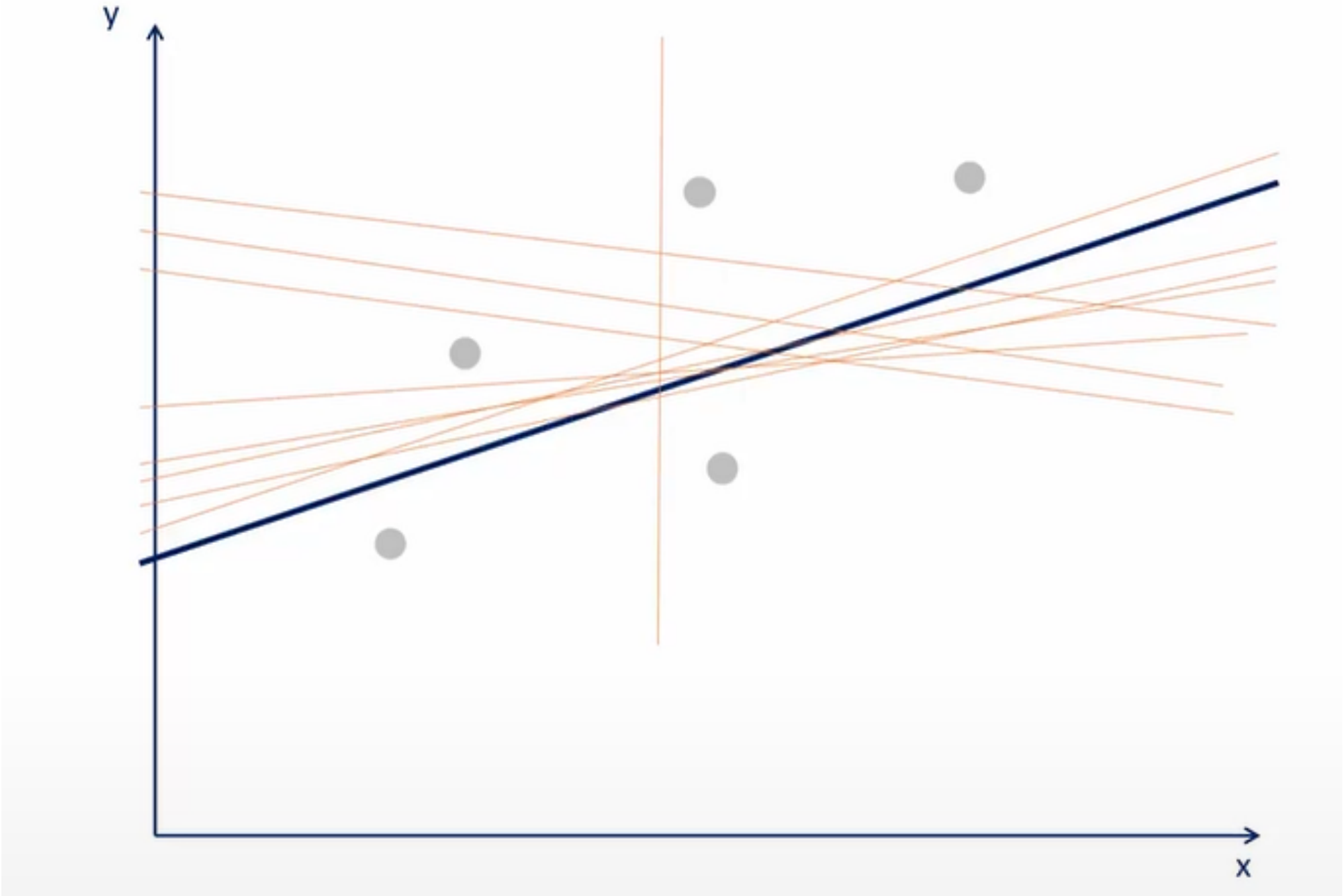


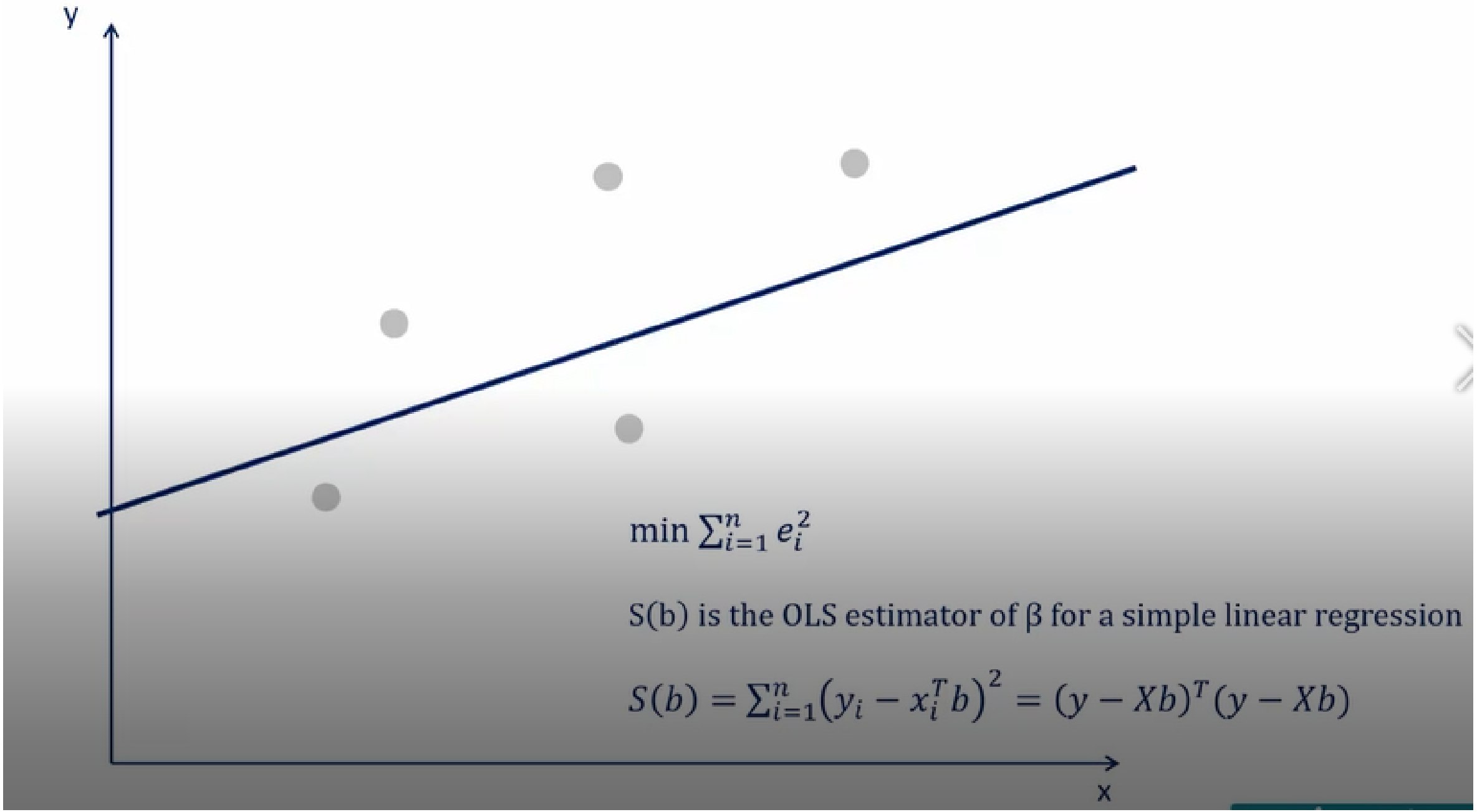
min **SSE**

Lower error => better explanatory power

LowEST error => BEST explanatory power


y
x

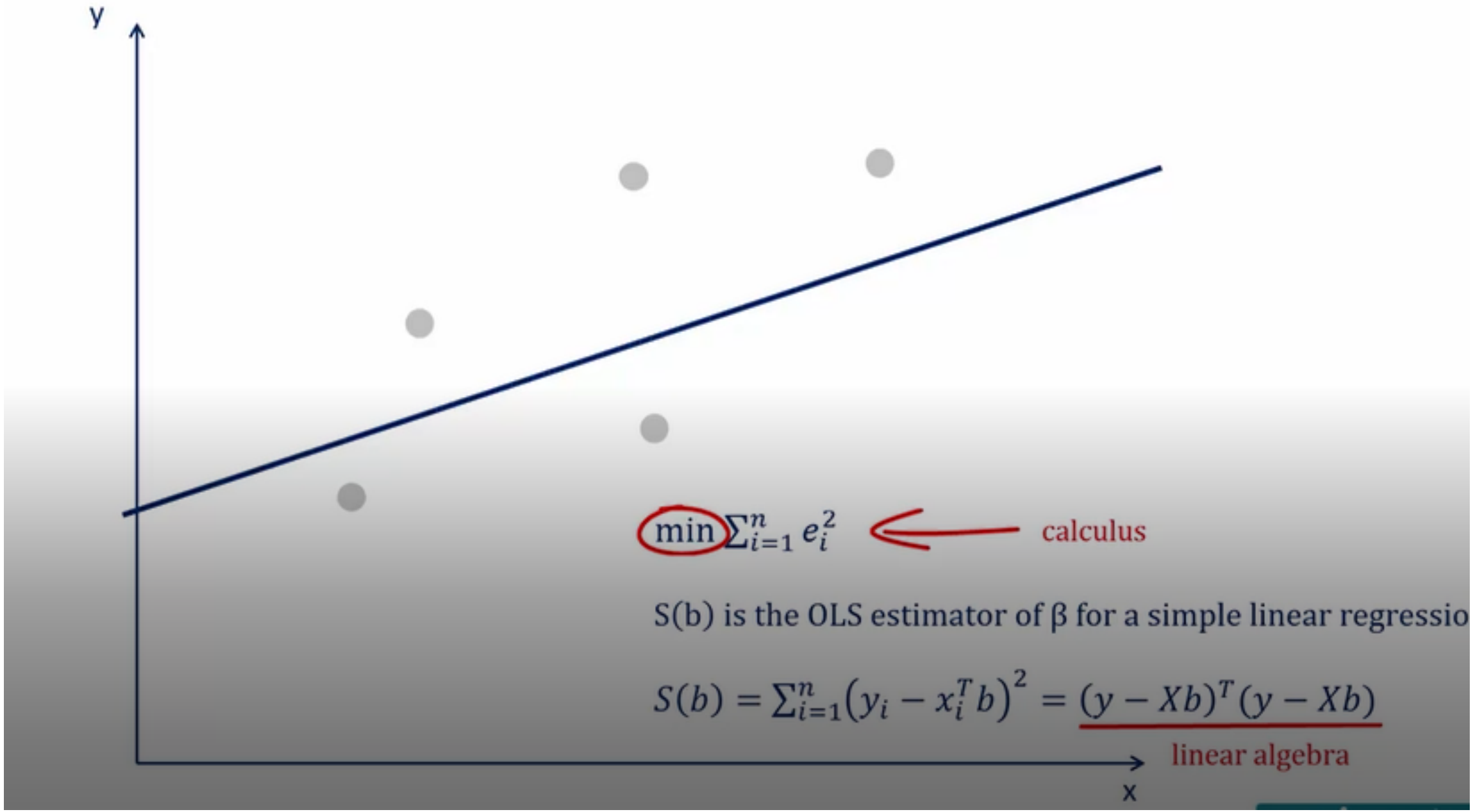

y
x

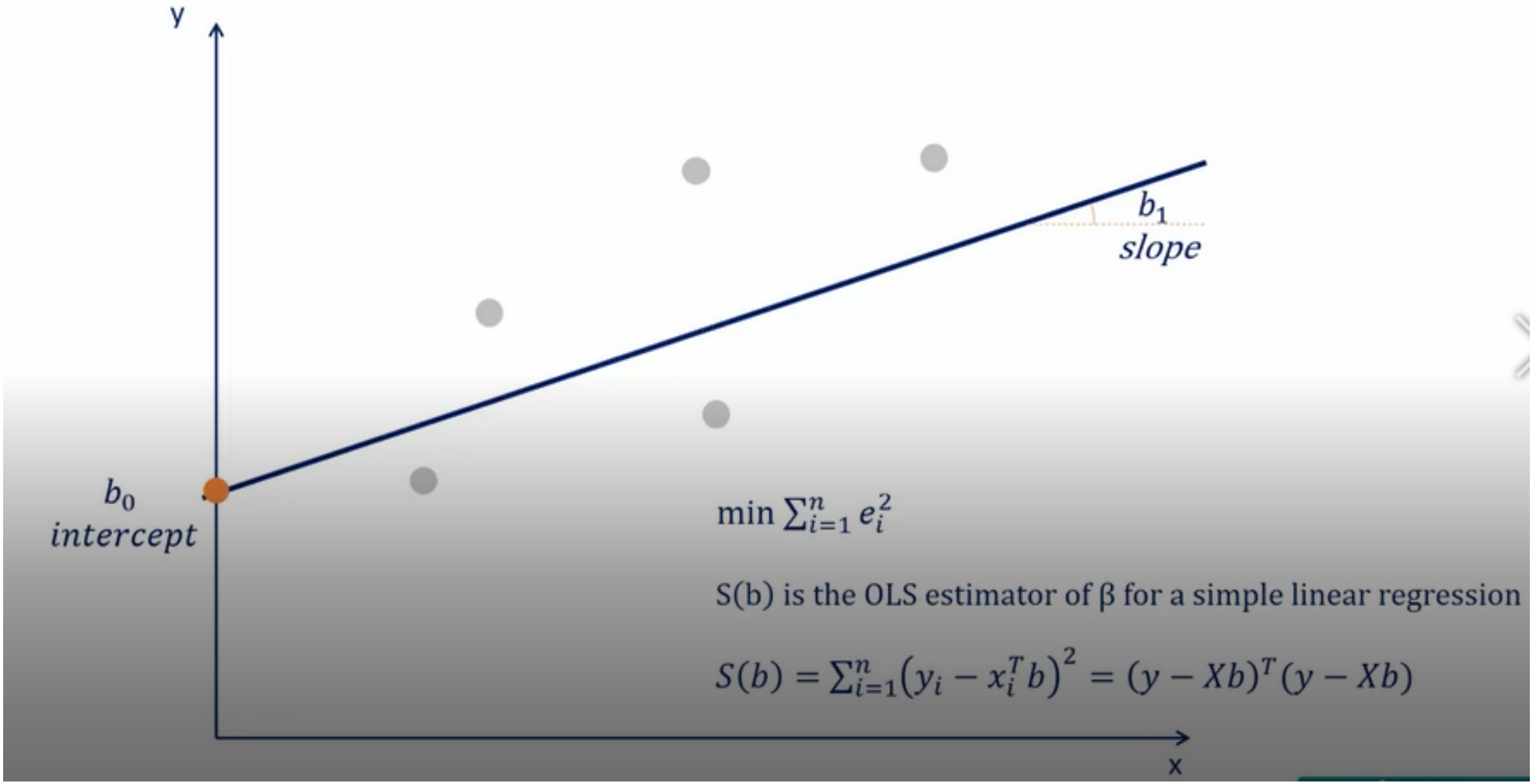


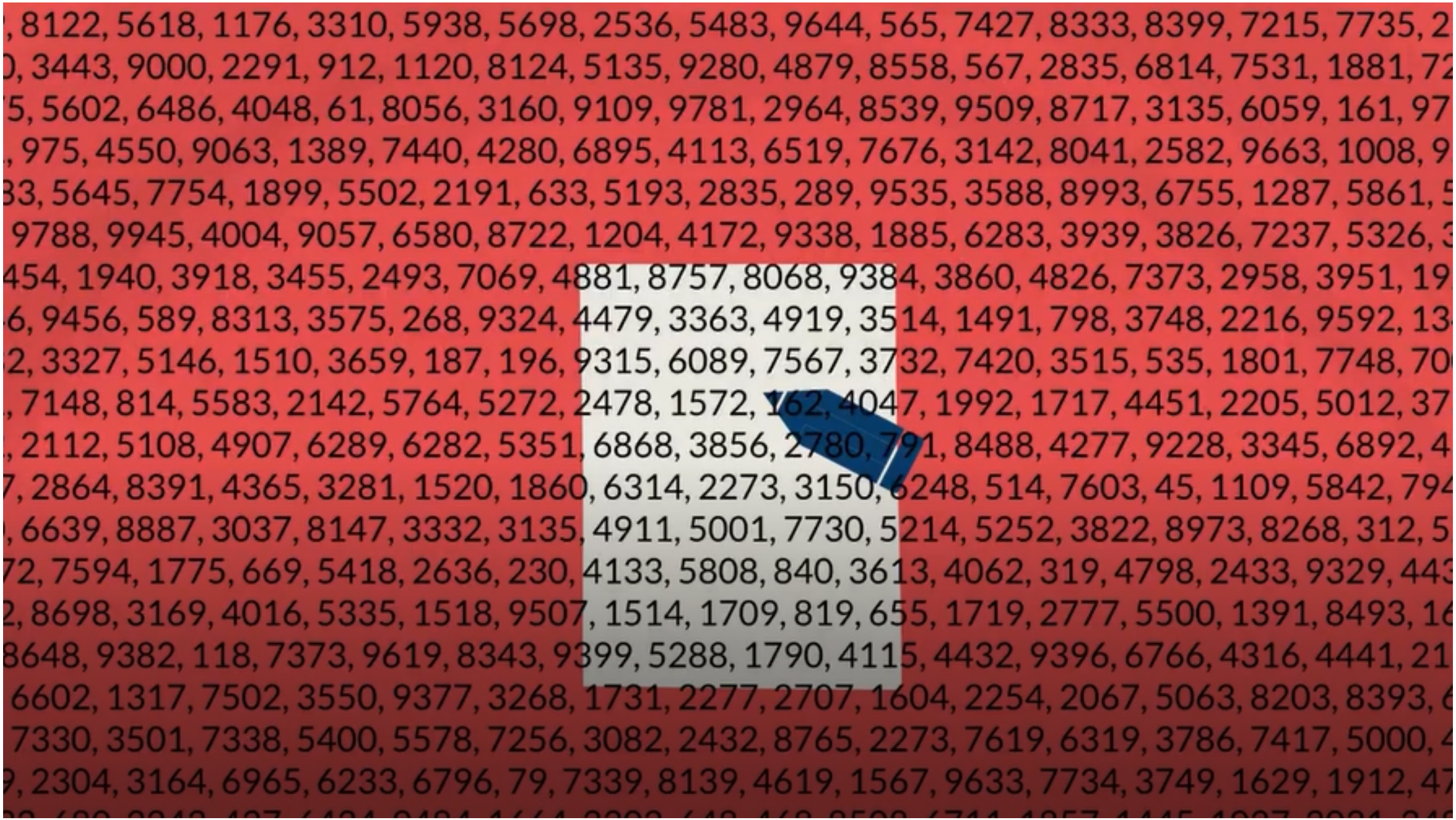
$\min \sum_{i=1}^{n} e_i^2$

S(b) is the OLS estimator of β for a simple linear regression

$$S(b) = \sum_{i=1}^{n} \left(y_i - x_i^T b\right)^2 = (y - Xb)^T (y - Xb)$$


y
x
$\min \sum_{i=1}^{n} e_i^2$
calculus
S(b) is the OLS estimator of β for a simple linear regressio
$S(b) = \sum_{i=1}^{n} (y_i - x_i^T b)^2 = (y - Xb)^T (y - Xb)$
linear algebra

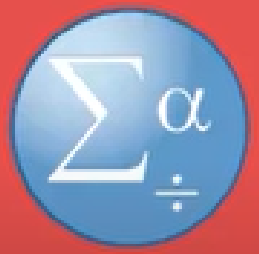

y
b_1
slope
b_0
intercept
$\min \sum_{i=1}^{n} e_i^2$
S(b) is the OLS estimator of β for a simple linear regression
$S(b) = \sum_{i=1}^{n} (y_i - x_i^T b)^2 = (y - Xb)^T (y - Xb)$
x


, 8122, 5618, 1176, 3310, 5938, 5698, 2536, 5483, 9644, 565, 7427, 8333, 8399, 7215, 7735, 2
0, 3443, 9000, 2291, 912, 1120, 8124, 5135, 9280, 4879, 8558, 567, 2835, 6814, 7531, 1881, 72
5, 5602, 6486, 4048, 61, 8056, 3160, 9109, 9781, 2964, 8539, 9509, 8717, 3135, 6059, 161, 97
, 975, 4550, 9063, 1389, 7440, 4280, 6895, 4113, 6519, 7676, 3142, 8041, 2582, 9663, 1008, 9
33, 5645, 7754, 1899, 5502, 2191, 633, 5193, 2835, 289, 9535, 3588, 8993, 6755, 1287, 5861, 5
9788, 9945, 4004, 9057, 6580, 8722, 1204, 4172, 9338, 1885, 6283, 3939, 3826, 7237, 5326, 3
454, 1940, 3918, 3455, 2493, 7069, 4881, 8757, 8068, 9384, 3860, 4826, 7373, 2958, 3951, 19
6, 9456, 589, 8313, 3575, 268, 9324, 4479, 3363, 4919, 3514, 1491, 798, 3748, 2216, 9592, 13
2, 3327, 5146, 1510, 3659, 187, 196, 9315, 6089, 7567, 3732, 7420, 3515, 535, 1801, 7748, 70
, 7148, 814, 5583, 2142, 5764, 5272, 2478, 1572, 162, 4047, 1992, 1717, 4451, 2205, 5012, 37
, 2112, 5108, 4907, 6289, 6282, 5351, 6868, 3856, 2780, 791, 8488, 4277, 9228, 3345, 6892, 4
7, 2864, 8391, 4365, 3281, 1520, 1860, 6314, 2273, 3150, 6248, 514, 7603, 45, 1109, 5842, 794
, 6639, 8887, 3037, 8147, 3332, 3135, 4911, 5001, 7730, 5214, 5252, 3822, 8973, 8268, 312, 5
72, 7594, 1775, 669, 5418, 2636, 230, 4133, 5808, 840, 3613, 4062, 319, 4798, 2433, 9329, 443
2, 8698, 3169, 4016, 5335, 1518, 9507, 1514, 1709, 819, 655, 1719, 2777, 5500, 1391, 8493, 16
8648, 9382, 118, 7373, 9619, 8343, 9399, 5288, 1790, 4115, 4432, 9396, 6766, 4316, 4441, 21
6602, 1317, 7502, 3550, 9377, 3268, 1731, 2277, 2707, 1604, 2254, 2067, 5063, 8203, 8393, 6
7330, 3501, 7338, 5400, 5578, 7256, 3082, 2432, 8765, 2273, 7619, 6319, 3786, 7417, 5000, 4
9, 2304, 3164, 6965, 6233, 6796, 79, 7339, 8139, 4619, 1567, 9633, 7734, 3749, 1629, 1912, 47

Software

Beginner to intermediate









Pro



